Shafick Kayemba

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Education

MASSACHUSETTS INSTITUTE OF TECHNOLOGY, Cambridge, MA

Bachelor of Science, Double Major in Computer Science & Engineering and Mathematics *Expected Feb 2027*

- **Relevant Coursework**: Distributed Systems, Database Systems, Operating Systems, Digital Systems, Computability and Complexity Theory, Computation Structures, Machine Learning
- **Expected Coursework**: Natural Language Processing

Experience

Software Engineering Intern June 2026 – August 2026

TikTok *Bellevue, WA*

- Evaluated multiple tree architectures and designed a query resolution tree that aggregates data from multiple sources, forming the core of a decision evaluation system serving over 10 downstream teams.
- Built a full-stack tree management platform using TypeScript and Go, integrating MySQL and a distributed configuration storage system to persist, retrieve, modify, and release decision trees; developed visualization tooling enabling engineering and QA teams to inspect, verify, and approve decision logic.
- Implemented versioned tree releases with optimistic concurrency control, ensuring consistency across configuration updates.

Software Engineering Intern June 2025 – August 2025

TikTok *San Jose, CA*

- Drove the design by assessing alternatives and trade-offs, selecting a message-queue database architecture, and implementing the ingestion framework to persist user event logs for future analytics.
- Shipped features to millions of users through gradual rollouts, leveraging feature flags, instrumentation, performance monitoring, and A/B testing to evaluate user behavior and guide iterations.
- Built an LLM-based automation system that applied prompt engineering and fine-tuning, implemented quantitative evaluation to score outputs, and integrated the model into the service.

Research Assistant June 2024 – June 2025

MIT Media Lab, Latent Lab *Cambridge, MA*

- Developed interactive visualization tools using UMAP and PCA to project high dimensional data embeddings into 2D space, enabling users to explore complex datasets and identify underlying relationships.
- Engineered a semantic axes feature that constructs user defined directions from natural language and projects high dimensional data onto these axes for interactive, interpretable analysis.
- Built a live visualization pipeline for file systems and Git repositories that automatically embeds newly added data, recomputes UMAP projections, and updates the 2D visualization in real time.

Data Engineering Intern June 2023 – August 2023

Triumvirate Environmental Inc. *Somerville, MA*

- Accelerated a mission-critical migration from a proprietary database to Oracle by using Python (Pandas) and SQL ETL scripts; delivered under tight deadlines, enabling a seamless transition and improved operational efficiency.
- Collaborated with 13 cross-functional teams to ensure data accuracy, completeness, and compliance with sustainability standards.

Lab Assistant August 2024 – December 2024

MIT Fundamentals of Programming *Cambridge, MA*

- Facilitated weekly programming labs and problem-solving sessions, guiding students through algorithm design, implementation, and alternative solution approaches.
- Provided hands-on debugging support by diagnosing code issues, explaining root causes, and helping students develop systematic testing and troubleshooting strategies.

Technical Skills

Languages: TypeScript, Python, C/C++, SQL, Go, Julia, MATLAB, HTML/CSS
Frameworks: React, Node.js, Flask, FastAPI
Databases: PostgreSQL, MySQL, MongoDB, ClickHouse, Hive
Developer Tools: Git, Docker, Supabase
Libraries: pandas, NumPy, Matplotlib, SciPy, scikit-learn, TensorFlow, PyTorch, pytest, cocotb